Top2018 Full Dataset — General Statistics and Geometric Analysis

Residues: 2,405,322 | **Chains:** 12,126 | **Structures:** 11,843 | **Source:** top2018full_measures.jsonl | **Generated by:** pydangle-biopython v0.5.1

This document provides the following analyses:

1. **General Information** — dataset overview, measurements available, and summary statistics
2. **Amino Acid Composition and Chain Length Distribution** — comparison with UniProt and PDB reference databases
3. **Torsion Angle Distributions** — univariate (backbone, omega deviation, sidechain) and multivariate (Ramachandran) distributions using circular statistics
4. **Bond Angle Distributions** — tau (N−Cα−C) and other non-periodic angles using linear statistics
5. **References**

1. General Information

Dataset	Top2018 Full
Total residues	2,405,322
Unique structures	11,843
Unique chains	12,126

Measurements available: phi: 2,027,343, psi: 2,027,390, omega: 2,027,346, tau: 2,405,312, chi1: 1,936,652, chi2: 1,417,035, chi3: 798,792, chi4: 362,877. Not all residues have all measurements — terminal residues lack phi/psi/omega, and glycine and alanine lack sidechain chi angles.

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Ramachandran category distribution:
  General      1,185,578  (49.29%)
  IleVal        242,781  (10.09%)
  Gly           151,712  ( 6.31%)
  TransPro       80,908  ( 3.36%)
  PrePro         63,787  ( 2.65%)
  CisPro          4,574  ( 0.19%)

Torsion angles (circular statistics):
  Angle             N    Circ Mean     Circ Std
  -------- ---------- ------------ ------------
  phi       2,027,343      -85.842       42.650
  psi       2,027,390       21.398      117.893
  omega     2,027,346      179.186        8.315
  chi1      1,936,652      -85.699       87.319
  chi2      1,417,035      158.075      125.702
  chi3        798,792     -173.409      102.217
  chi4        362,877       -1.809       83.632

Bond angles (linear statistics):
  tau       2,405,312      111.153        2.531
```

2. Amino Acid Composition and Chain Length Distribution

2.1 Representativeness

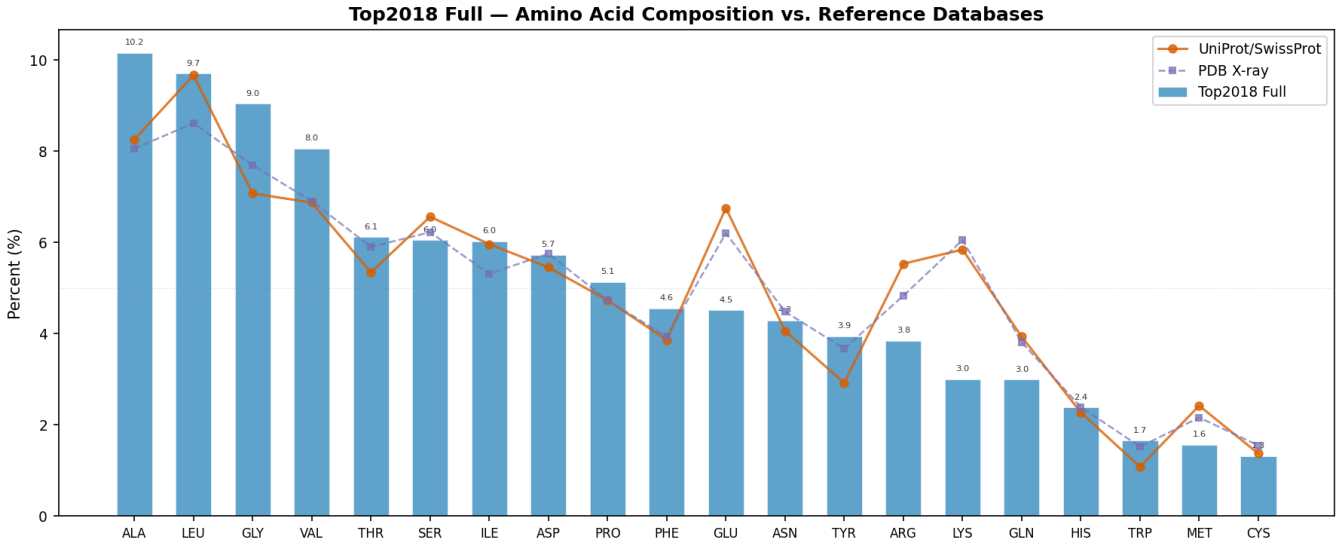
The figure and table below compare the Top2018 Full amino acid frequencies against two reference databases:

- **UniProt/SwissProt** (~2024): amino acid frequencies across all reviewed protein sequences, representing the broadest available view of protein sequence space.
- **PDB X-ray** (~2025): frequencies from a 10,000-entity sample of X-ray crystallographic structures in the RCSB Protein Data Bank.

The two reference distributions are highly similar (most amino acids differ by <0.5%), indicating that the PDB's well-known crystallization bias — overrepresentation of soluble, globular, well-expressing proteins from model organisms — has only a modest effect on overall amino acid composition.

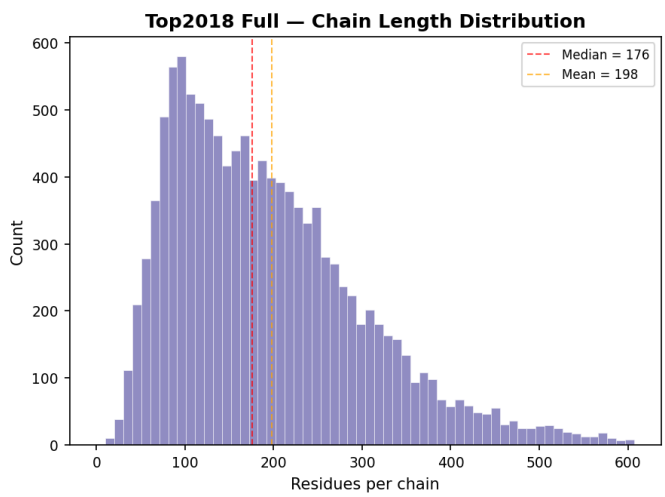
The Top2018 Full dataset (2,405,322 residues from 11,843 structures) shows additional deviations from both references due to quality filtering, homology reduction, and sample size effects.

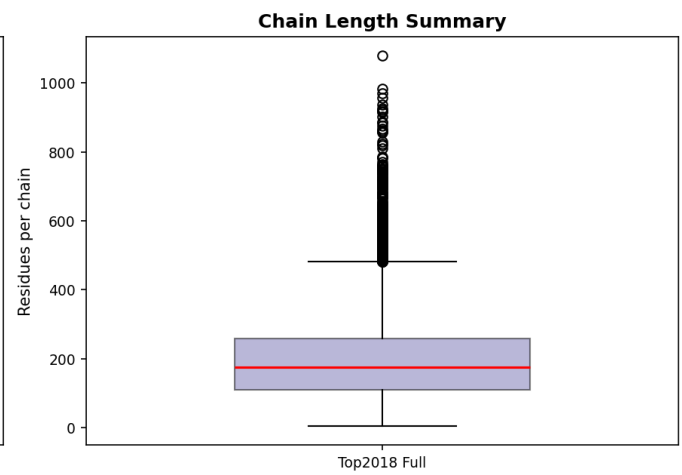
Note: This dataset shows substantial deviations from reference amino acid frequencies (e.g., LYS (-2.8%); GLU (-2.2%); GLY (+2.0%)). This is expected for datasets with stringent quality filtering, which preferentially retains well-ordered core residues (enriching Ala, Val, Gly) and removes disordered surface residues (depleting Lys, Glu, Arg). A detailed analysis of these filtering effects is deferred to future work.



Residue	Count	Dataset%	UniProt%	PDB%	Δ UniProt
ALA	244,156	10.15%	8.25%	8.05%	+1.90%
LEU	233,250	9.70%	9.66%	8.61%	+0.04%
GLY	217,285	9.03%	7.07%	7.69%	+1.96%
VAL	193,595	8.05%	6.87%	6.90%	+1.18%
THR	147,063	6.11%	5.34%	5.90%	+0.77%
SER	145,478	6.05%	6.56%	6.22%	-0.51%
ILE	144,935	6.03%	5.96%	5.31%	+0.07%
ASP	137,763	5.73%	5.45%	5.75%	+0.28%
PRO	123,406	5.13%	4.73%	4.73%	+0.40%
PHE	109,593	4.56%	3.86%	3.93%	+0.70%
GLU	108,437	4.51%	6.75%	6.20%	-2.24%
ASN	103,009	4.28%	4.06%	4.48%	+0.22%
TYR	94,688	3.94%	2.92%	3.67%	+1.02%
ARG	92,343	3.84%	5.53%	4.83%	-1.69%
LYS	72,214	3.00%	5.84%	6.05%	-2.84%
GLN	71,950	2.99%	3.93%	3.81%	-0.94%
HIS	57,177	2.38%	2.27%	2.39%	+0.11%
TRP	39,718	1.65%	1.08%	1.52%	+0.57%
MET	37,543	1.56%	2.42%	2.16%	-0.86%
CYS	31,719	1.32%	1.37%	1.55%	-0.05%

2.2 Chain Length Distribution



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Chain length statistics (12,126 chains):
  Min:              4
  Q1:             109
  Median:         176
  Mean:         198.4
  Q3:             258
  Max:          1,080
  Std:          119.5
```
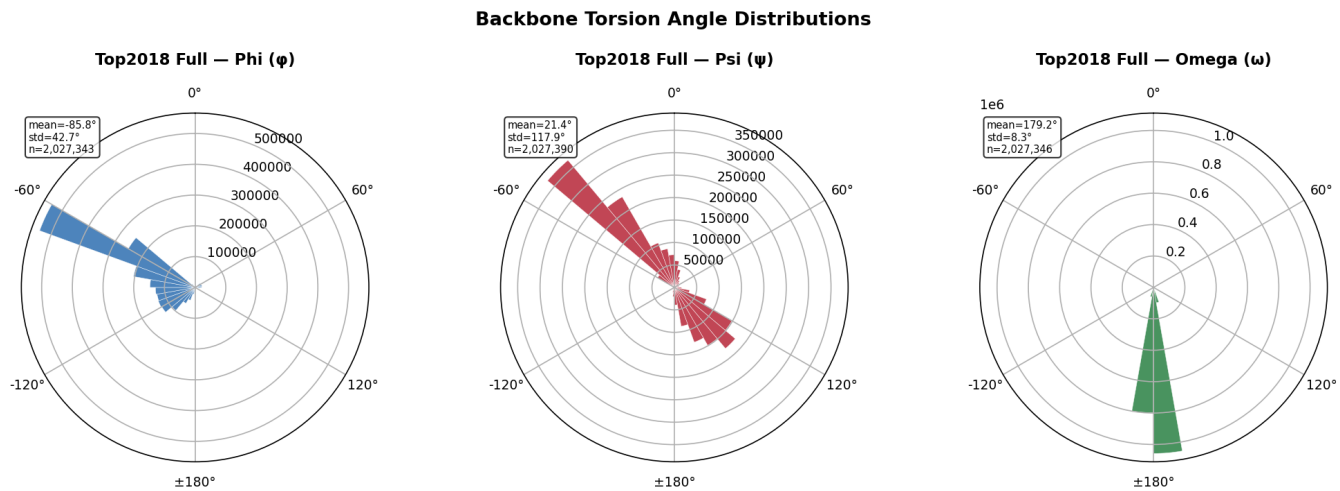

3. Torsion Angle Distributions (Circular Statistics)

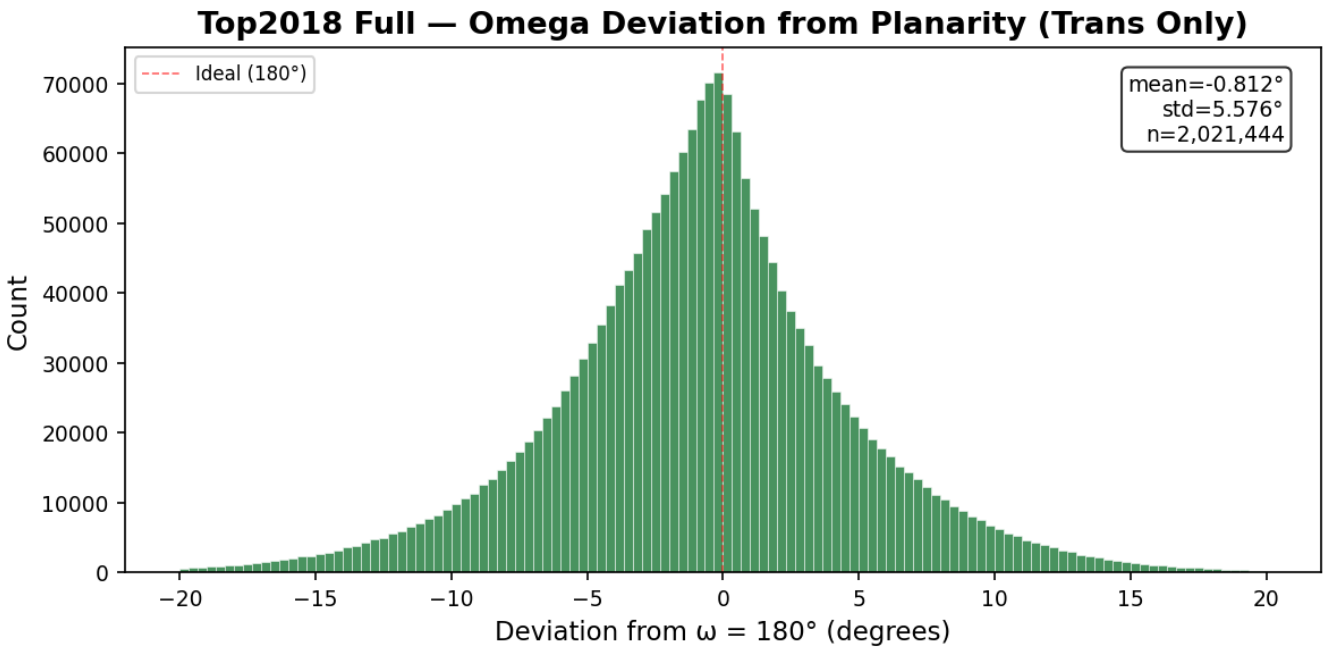
Torsion angles (φ, ψ, ω, χ) are periodic variables defined on the interval [−180°, +180°]. Standard (linear) arithmetic means and standard deviations give misleading results when values cluster near the ±180° boundary — for example, a linear mean of trans peptide bond ω values near +179° and −179° yields ~0° rather than the correct ~180°. All torsion angle summary statistics in this report use **circular statistics**: the circular mean is computed as atan2(⟨sin θ⟩, ⟨cos θ⟩), and the circular standard deviation as √(−2 ln R̄) where R̄ is the mean resultant length.

3.1 Univariate Distributions

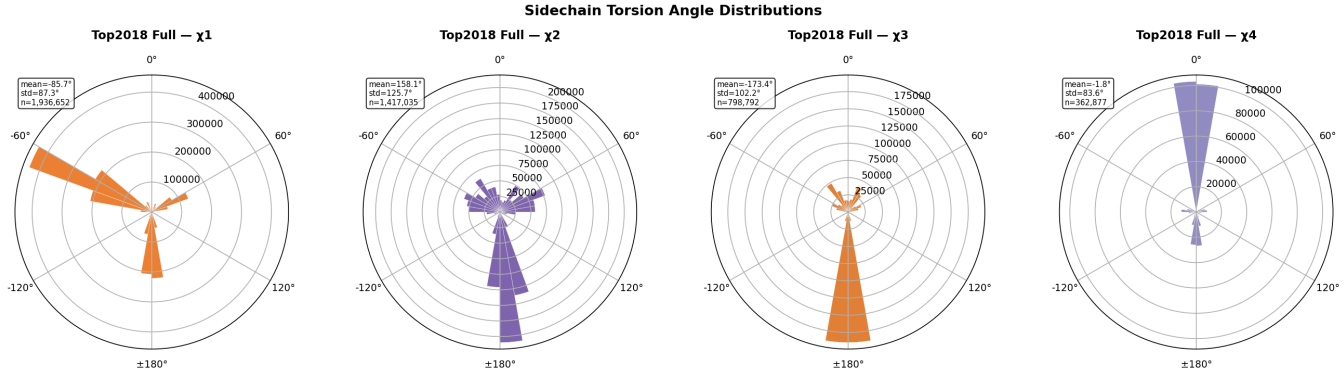
3.1.1 Backbone Torsion Angles (φ, ψ, ω)



3.1.2 Omega Deviation from Planarity (Trans Peptides Only)

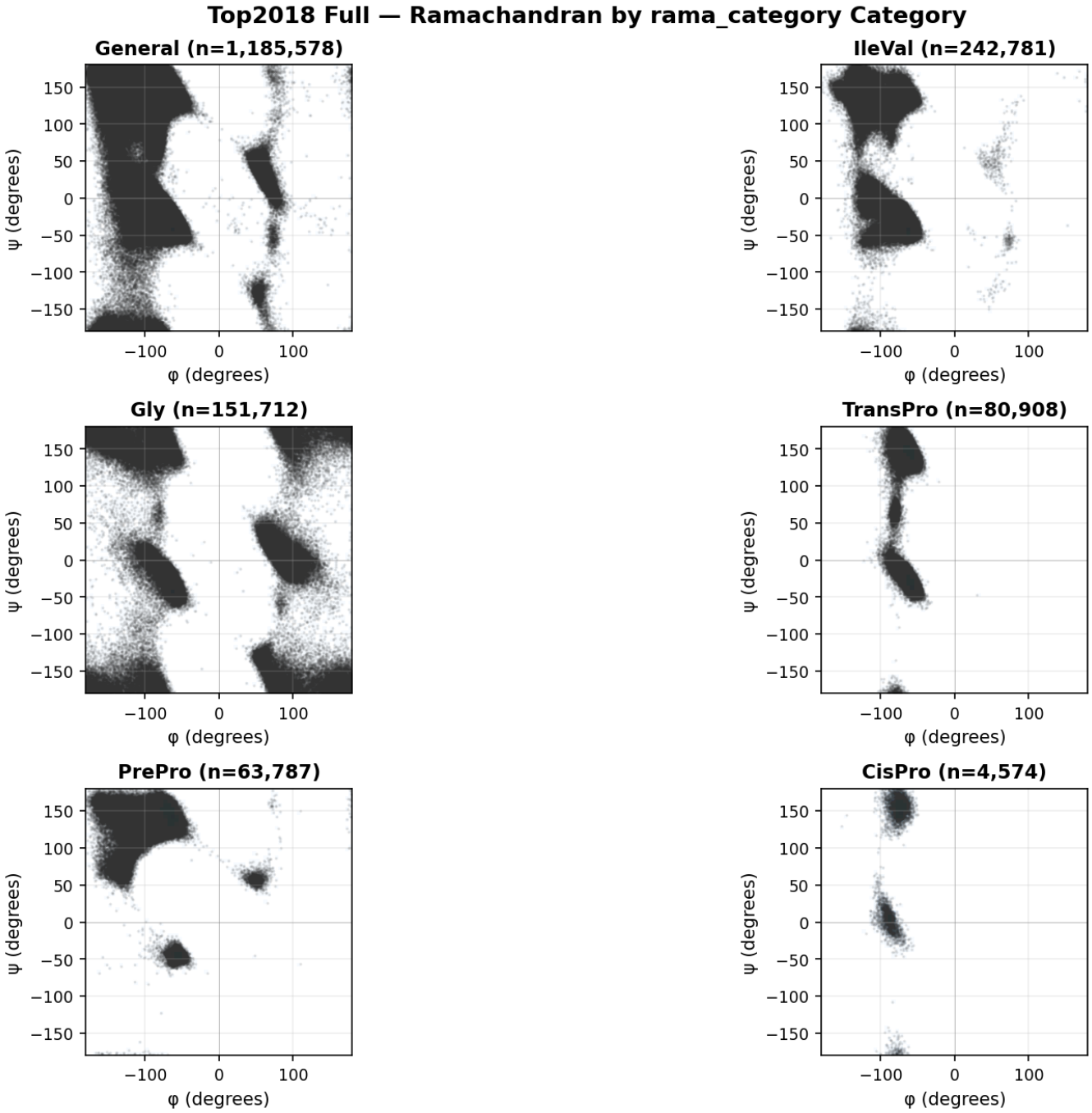


3.1.3 Sidechain Torsion Angles (χ1–χ4)



3.2 Multivariate Distributions

3.2.1 Ramachandran Distributions (φ × ψ by Category)

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Category counts (rama_category):
  General      1,185,578  (49.29%)
  IleVal         242,781  (10.09%)
  Gly            151,712  ( 6.31%)
  TransPro        80,908  ( 3.36%)
  PrePro          63,787  ( 2.65%)
  CisPro           4,574  ( 0.19%)
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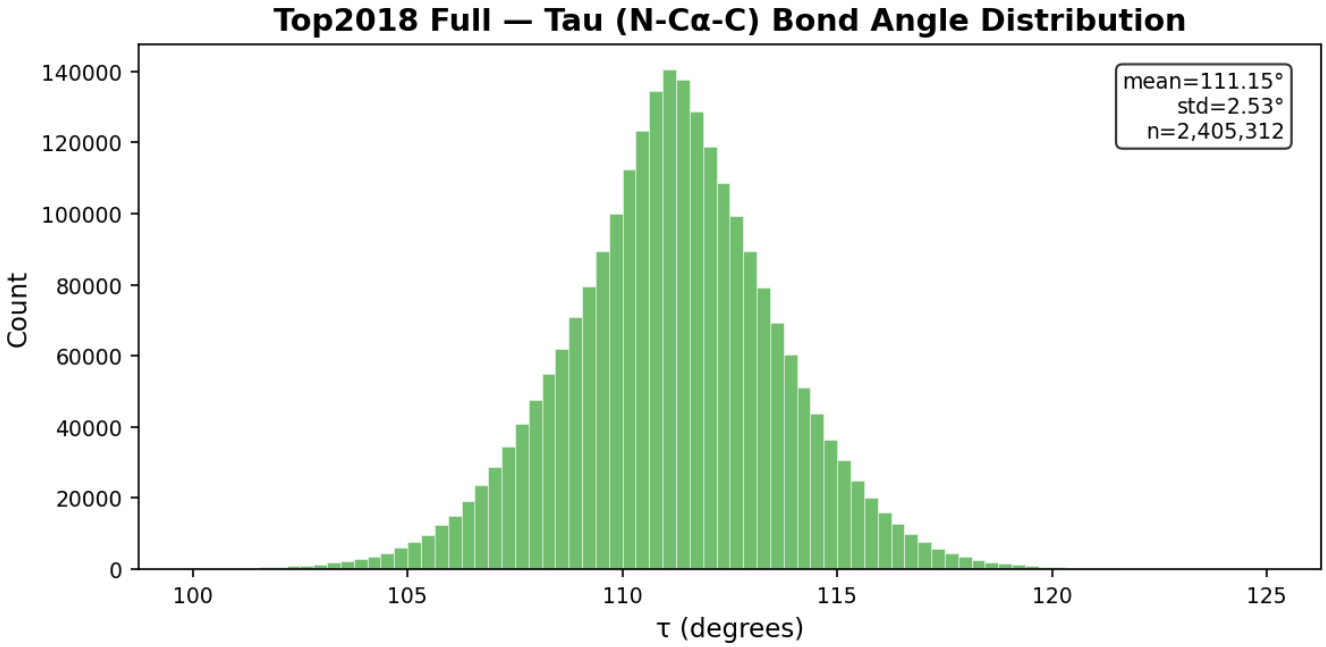
4. Bond Angle Distributions (Linear Statistics)

Bond angles are non-periodic and are analyzed with standard linear statistics (arithmetic mean, standard deviation).

4.1 Tau (N−Cα−C)



5. References

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